

States–Japan–United Kingdom frontiers lie inside the G5 frontier. There is still a benefit in adding Germany and France from the United States–Japan–United Kingdom, but it is not as big a benefit as adding the United Kingdom starting from the United States–Japan. That is, although Happy Buddha continues getting happier by adding countries, the rate at which he becomes happier decreases. There are *decreasing marginal diversification benefits* as we add assets. As we continue adding assets beyond the G5, the frontier will continue to expand but the added diversification benefits become smaller.

Figures 3.4 and 3.5 show that if we add an asset, the mean-variance frontier gets fatter. Conversely, if we remove an asset, the mean-variance frontier shrinks. Removing assets, as Norway did by divesting Wal-Mart, can only cause the maximum Sharpe ratio to (weakly) decrease. Thus, constraining the fund manager's portfolio by throwing out Norway has decreased investment opportunities and lowered the maximum achievable risk-return trade-off. We will later compute how the mean-variance frontier shrinks as we remove assets.

The mathematical statement of the problem in Figures 3.4 and 3.5 is:

$$\begin{aligned} & \min_{\{w_i\}} \text{var}(r_p) \\ & \text{subject to } E(r_p) = \mu^* \text{ and } \sum_i w_i = 1, \end{aligned} \quad (3.2)$$

where the portfolio weight for asset i is w_i . We find the combination of portfolio weights, $\{w_i\}$, that minimizes the portfolio variance subject to two constraints. The first is that the expected return on the portfolio is equal to a target return, μ^* . The second is that the portfolio must be a valid portfolio, which is the admissibility condition that we have seen earlier. Students of operations research will recognize equation (3.2) as a *quadratic programming problem*, and what makes mean-variance investing powerful (but alas, misused; see below) is that there are very fast algorithms for solving these types of problems.

Figure 3.6 shows how this works pictorially. Choosing a target return of $\mu^* = 10\%$, we find the portfolio with the lowest volatility (or variance). We plot this with an X. Then we change the target return to $\mu^* = 12\%$. Again we find the portfolio with the lowest volatility and plot this with another X. The mean-variance frontier is drawn by changing the target return, μ^* and then linking all the Xs for each target return. Thus, the mean-variance frontier is a *locus* of points, where each point denotes the minimum variance achievable for each expected return.

2.4. CONSTRAINED MEAN-VARIANCE FRONTIERS

So far, we have constructed unconstrained mean-variance frontiers. But investors often face constraints on what types of portfolios that they can hold. One constraint faced by many investors is the inability to short. When there is a *no short-sales constraint*, all the portfolio weights have to be positive ($|w_i| \geq 0$), and we can add this constraint to the optimization problem in equation (3.2).